University of Ottawa

Department of Mathematics and Statistics

MAT 1302C: Mathematical Methods II
Instructor: Hadi Salmasian

Midterm 1 – Group A
October 10, 2013

Surname _____ First Name _____

Student # _____ DGD _____

Instructions:

(1) You have 80 minutes to complete this exam.
(2) Show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.
(3) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
provided.
(4) No notes, books, calculators or scrap paper are allowed.
(5) The final page of the exam may be used for scrap work.
(6) Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	3	4	2	4	2	5	20
Score							

1. **(3 points)** Determine if the vector equation

$$x_1\begin{bmatrix}1\\2\\1\end{bmatrix}+x_2\begin{bmatrix}-2\\-4\\-2\end{bmatrix}+x_3\begin{bmatrix}-1\\1\\2\end{bmatrix}+x_4\begin{bmatrix}3\\0\\-3\end{bmatrix}=\begin{bmatrix}1\\5\\4\end{bmatrix}$$

is consistent or inconsistent. If it is consistent, determine its general solution.

Solution:

The augmented matrix associated to the system corresponding to the above vector equation is

$$\left[\begin{array}{cccc|c}1&-2&-1&3&1\\2&-4&1&0&5\\1&-2&2&-3&4\end{array}\right]$$

It is row reduced as follows:

$$\left[\begin{array}{cccc|c}1&-2&-1&3&1\\2&-4&1&0&5\\1&-2&2&-3&4\end{array}\right]\xrightarrow[R_2\to R_2-2R_1]{R_3\to R_3-R_1}\left[\begin{array}{cccc|c}1&-2&-1&3&1\\0&0&3&-6&3\\0&0&3&-6&3\end{array}\right]\xrightarrow{R_3\to R_3-R_2}\left[\begin{array}{cccc|c}1&-2&-1&3&1\\0&0&3&-6&3\\0&0&0&0&0\end{array}\right]\xrightarrow{R_2\to\frac{1}{3}R_2}\left[\begin{array}{cccc|c}1&-2&-1&3&1\\0&0&1&-2&1\\0&0&0&0&0\end{array}\right]\xrightarrow{R_1\to R_1+R_2}\left[\begin{array}{cccc|c}1&-2&0&1&2\\0&0&1&-2&1\\0&0&0&0&0\end{array}\right]$$

The rightmost column is not a pivot column, so the system is consistent. Moreover, x_2 and x_4 are free variables, and x_1, x_3 are basic variables. One obtains $\begin{cases}x_1=2-x_4+2x_2\\x_2=\text{free}\\x_3=2x_4+1\\x_4=\text{free}\end{cases}$

2. **(4 points)** Is the following linear system consistent? If so, find the general solution and write it in **vector parametric form**.

$$\begin{array}{rcrcrcrcl} x & + & 2y & - & 2z & + & 3w & = & 2 \\ 2x & + & 4y & - & 3z & + & 4w & = & 5 \\ 5x & + & 10y & - & 8z & + & 11w & = & 12 \end{array}$$

Solution: The augmented matrix of the above system of linear equations is

$$A = \left[\begin{array}{cccc|c} 1 & 2 & -2 & 3 & 2 \\ 2 & 4 & -3 & 4 & 5 \\ 5 & 10 & -8 & 11 & 12 \end{array}\right]$$

It is row reduced as follows:

$$A = \left[\begin{array}{cccc|c} 1 & 2 & -2 & 3 & 2 \\ 2 & 4 & -3 & 4 & 5 \\ 5 & 10 & -8 & 11 & 12 \end{array}\right] \xrightarrow[R_2 \to R_2 - 2R_1]{R_3 \to R_3 - 5R_1} \left[\begin{array}{cccc|c} 1 & 2 & -2 & 3 & 2 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 2 & -4 & 2 \end{array}\right] \xrightarrow{R_3 \to R_3 - 2R_2} \left[\begin{array}{cccc|c} 1 & 2 & -2 & 3 & 2 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to R_1 + 2R_2} \left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 4 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The rightmost column is not a pivot column, therefore the system is consistent, and moreover: x, z are basic variables and y, w are free. It follows that $\begin{cases} x = -2y + w + 4 \\ y = \text{free} \\ z = 2w + 1 \\ w = \text{free} \end{cases}$

and the vector parametric form of the solution is:

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -2y + w + 4 \\ y \\ 2w + 1 \\ w \end{bmatrix} = y \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + w \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

3. **(2 points)** Find <u>all</u> of the value(s) of h for which the following system has a unique solution:

$$\begin{array}{rcrcl} x & + & y & = & 7 \\ -9x & + & hy & = & 8 \end{array}$$

Solution:
The augmented matrix of the above system of linear equations is

$$A = \left[\begin{array}{cc|c} 1 & 1 & 7 \\ -9 & h & 8 \end{array}\right] \xrightarrow{R_2 \to R_2 + 9R_1} \left[\begin{array}{cc|c} 1 & 1 & 7 \\ 0 & h+9 & 71 \end{array}\right]$$

To get a unique solution, one must impose that $h \neq -9$.

4. **(4 points)** Suppose that $\mathbf{v}_1 = \begin{bmatrix}1\\0\\1\\2\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}2\\1\\1\\2\end{bmatrix}$. Is the vector $\mathbf{b} = \begin{bmatrix}4\\3\\1\\2\end{bmatrix}$ in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$? Justify your answer!

Solution:

Note that the augmented matrix of the corresponding system can be row reduced as follows:

$$\left[\begin{array}{ccc|c}1&1&2&4\\0&1&1&3\\1&1&1&1\\2&1&2&2\end{array}\right] \xrightarrow[R_4 \to R_4 - 2R_1]{R_3 \to R_3 - R_1} \left[\begin{array}{ccc|c}1&1&2&4\\0&1&1&3\\0&0&-1&-3\\0&-1&-2&-6\end{array}\right] \xrightarrow{R_4 \to R_4 + R_2} \left[\begin{array}{ccc|c}1&1&2&4\\0&1&1&3\\0&0&-1&-3\\0&0&-1&-3\end{array}\right] \xrightarrow{R_4 \to R_4 - R_3} \left[\begin{array}{ccc|c}1&1&2&4\\0&1&1&3\\0&0&-1&-3\\0&0&0&0\end{array}\right] \xrightarrow{R_3 \to (-1)R_3}$$

$$\left[\begin{array}{ccc|c}1&1&2&4\\0&1&1&3\\0&0&1&3\\0&0&0&0\end{array}\right] \xrightarrow[R_1 \to R_1 - 2R_3]{R_2 \to R_2 - R_3} \left[\begin{array}{ccc|c}1&1&0&-2\\0&1&0&0\\0&0&1&3\\0&0&0&0\end{array}\right] \xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|c}1&0&0&-2\\0&1&0&0\\0&0&1&3\\0&0&0&0\end{array}\right]$$

Because the system is consistent, the answer is positive.

5. **(2 points)** For what value(s) of h are the following vectors linearly independent? Justify your answer!

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \\ 4 \\ -3 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ -1 \\ -3 \\ h \end{bmatrix}$$

Solution:

Note that

$$\begin{bmatrix} 1 & -1 & 1 \\ 3 & 2 & -1 \\ 1 & 4 & -3 \\ 3 & -3 & h \end{bmatrix} \xrightarrow[R_4 \to R_4 - 3R_1]{\substack{R_2 \to R_2 - 3R_1 \\ R_3 \to R_3 - R_1}}$$

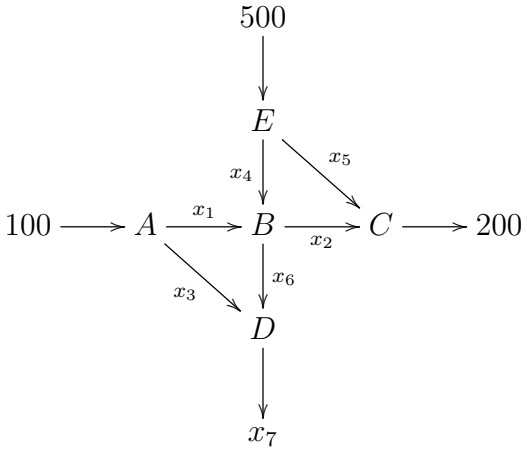
$$\to \begin{bmatrix} 1 & -1 & 1 \\ 0 & 5 & -4 \\ 0 & 5 & -4 \\ 0 & 0 & h-3 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2}$$

$$\to \begin{bmatrix} 1 & -1 & 1 \\ 0 & 5 & -4 \\ 0 & 0 & 0 \\ 0 & 0 & h-3 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_4}$$

$$\to \begin{bmatrix} 1 & -1 & 1 \\ 0 & 5 & -4 \\ 0 & 0 & h-3 \\ 0 & 0 & 0 \end{bmatrix}$$

When $h \neq 3$, every column will be a pivot column, therefore the vectors will be linearly independent.

6. **(5 points)** Consider the traffic flow described by the following diagram. The letters A through E label intersections and numerical values indicate flow in cars per minute. The arrows indicate the direction of flow (all roads are one-way).



(a) Write down a linear system describing the traffic flow, as well as all constraints on the variables $x_i, i = 1, \ldots, 7$. (Do not solve the linear system at this stage.)

Solution:

A: $100 = x_1 + x_3$

B: $x_1 + x_4 = x_2 + x_6$

C: $x_5 + x_2 = 200$

D: $x_3 + x_6 = x_7$

E: $500 = x_4 + x_5$

Total: $100 + 500 = 200 + x_7$

(b) The reduced echelon form of the augmented matrix corresponding to the linear system in part (a) is:

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & 0 & -1 & 0 & -300 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 200 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 400 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 500 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 400 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Find the general flow pattern.

Solution: x_5 and x_6 are free variables; while the rest of variables are basic. The general solution is:

$$\begin{cases} x_1 = -300 + x_6 \\ x_2 = 200 - x_5 \\ x_3 = 400 - x_6 \\ x_4 = 500 - x_5 \\ x_5 = \text{free} \\ x_6 = \text{free} \\ x_7 = 400 \end{cases}$$

(c) Assume that due to road work, the flow along EC is limited to a maximum of 50 cars per minute. Find the minimum flow along BC.

Solution:
(c) Since $x_5 \leq 50$ and $x_2 = 200 - x_5$ one gets that $x_2 \geq 200 - 50 = 150$.

This page is for scrap work and is intentionally left blank.